STACKS, DISCRIMINANTS: LEMMA-TAU-NONZERO

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1126-1136 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 1138-1159, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 30 of queue_discriminant.json (pb=882, stmt 1126-1136, proof 1138-1159).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00J3: sources/stacks-discriminant/pinned-original-source.tex, source lines 1159–1159. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BIL: sources/stacks-discriminant/pinned-original-source.tex, source lines 1159–1159. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BT8: sources/stacks-discriminant/pinned-original-source.tex, source lines 874–877. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let k be a field and let A be a finite k-algebra. Assume A is local with residue field k'. The following are equivalent*

(1) *$\mathrm{Trace}_{A/k}$ is nonzero,*
(2) *$\tau_{A/k} \in \omega_{A/k}$ is nonzero, and*
(3) *k'/k is separable and $\mathrm{length}_A(A)$ is prime to the characteristic of k.*

Proof. Conditions (1) and (2) are equivalent by Lemma 0BT8. Let $\mathfrak{m} \subset A$. Since $\dim_k(A) < \infty$ it is clear that A has finite length over A. Choose a filtration

$$A = I_0 \supset \mathfrak{m} = I_1 \supset I_2 \supset \ldots I_n = 0$$

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by ideals such that $I_i/I_{i+1} \cong k'$ as A-modules. See Algebra, Lemma 00J3 which also shows that $n = \text{length}_A(A)$. If $a \in \mathfrak{m}$ then $aI_i \subset I_{i+1}$ and it is immediate that $\text{Trace}_{A/k}(a) = 0$. If $a \notin \mathfrak{m}$ with image $\lambda \in k'$, then we conclude

$$\text{Trace}_{A/k}(a) = \sum\nolimits_{i=0,\ldots,n-1} \text{Trace}_k(a : I_i/I_{i-1} \to I_i/I_{i-1}) = n\text{Trace}_{k'/k}(\lambda)$$

The proof of the lemma is finished by applying Fields, Lemma 0BIL. $\square$